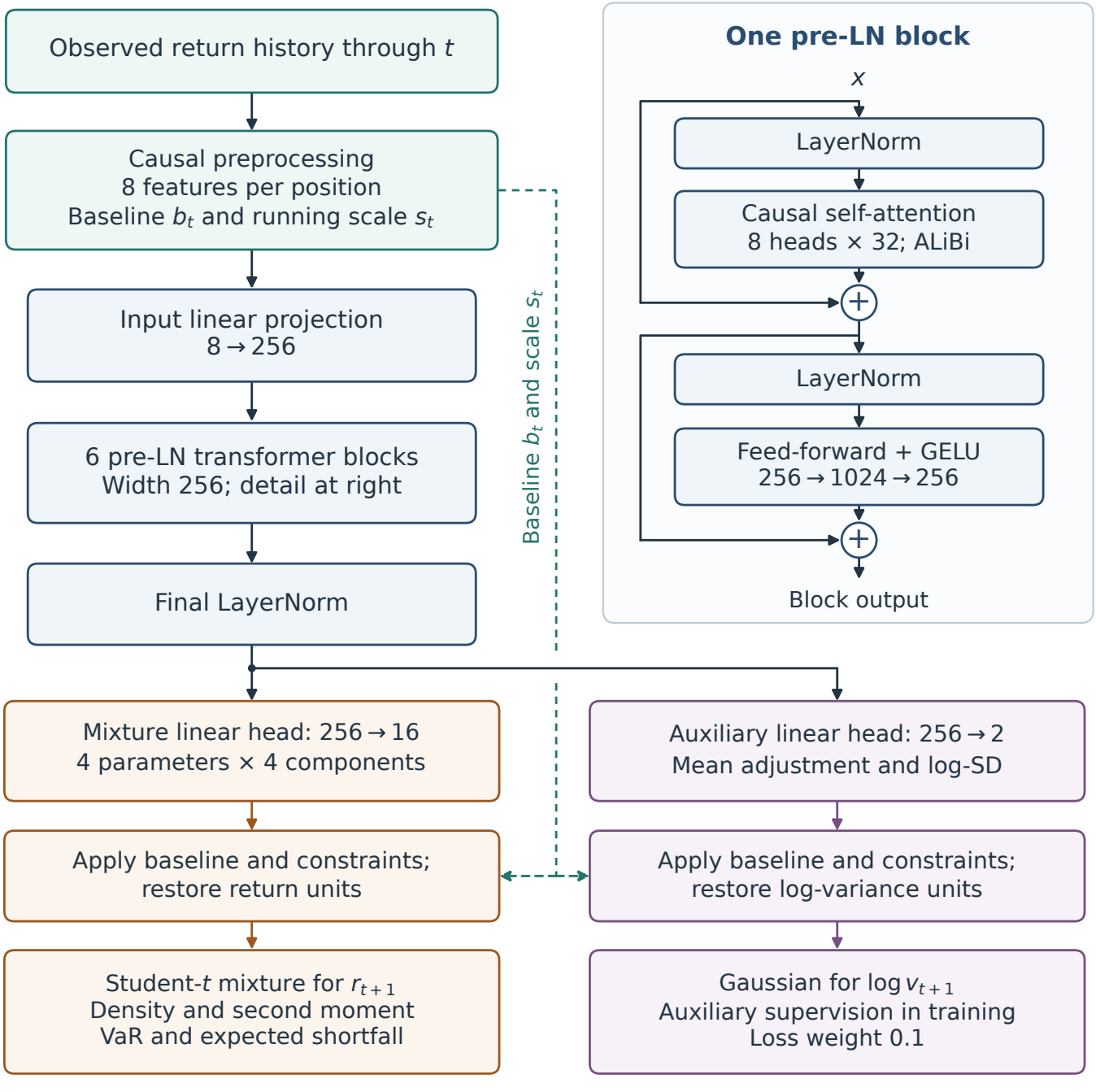
Observed return history through t
Causal preprocessing
8 features per position
Baseline b_t and running scale s_t
Input linear projection
$8 \to 256$
6 pre-LN transformer blocks
Width 256; detail at right
Final LayerNorm
Baseline b_t and scale s_t
One pre-LN block
x
LayerNorm
Causal self-attention
8 heads × 32; ALiBi
LayerNorm
Feed-forward + GELU
$256 \to 1024 \to 256$
Block output
Mixture linear head: $256 \to 16$
4 parameters × 4 components
Auxiliary linear head: $256 \to 2$
Mean adjustment and log-SD
Apply baseline and constraints;
restore return units
Apply baseline and constraints;
restore log-variance units
Student-t mixture for r_{t+1}
Density and second moment
VaR and expected shortfall
Gaussian for $\log v_{t+1}$
Auxiliary supervision in training
Loss weight 0.1